A STATISTICAL STUDY OF LIVE BIRTH AND CRUDE BIRTH OVER THE YEARS OF KERALA

BY TONY THOMAS

MSC STATISTICS

ST. ALBERTS COLLEGE, ERNAKULAM

CHAPTER 1

INTRODUCTION

This study focuses on the analysis of live births and crude birth rate trends in Kerala over the period 1963 to 2023, covering six decades of demographic transition. Live births represent the total number of births occurring in a given year, while the crude birth rate (CBR) measures the number of live births per 1,000 population, serving as a key indicator of population growth.

Over the years, Kerala has experienced significant socio-economic and demographic changes, including improvements in education, healthcare, and family planning practices. These factors have contributed to notable shifts in birth patterns, making it important to examine long-term trends and variations in birth rates.

The objective of this study is to:

- Analyze the temporal trend of live births and crude birth rate
- Identify patterns and structural changes over time
- Understand the impact of demographic transition on birth behaviour
- Provide a basis for forecasting future birth trends

This analysis uses statistical and graphical methods to explore the data, offering insights that are relevant for policy planning, resource allocation, and understanding population dynamics in Kerala.

Methodology

This study adopts a time series analytical framework to examine the trends in live births and crude birth rate in Kerala over the period 1963–2023. The methodology combines descriptive statistics, graphical analysis, and forecasting techniques to understand both historical patterns and future projections.

Initially, the dataset was organized and cleaned to ensure consistency in time series format. Exploratory Data Analysis (EDA) was carried out using line graphs and summary statistics to identify long-term trends and variations in live births and crude birth rate.

To capture short-term fluctuations, year-on-year percentage change was computed, providing insights into the rate of increase or decrease in fertility indicators across consecutive years. Additionally, heatmap visualization was employed to represent the intensity of crude birth rate over time, enabling clear identification of periods of high and low fertility.

For formal time series modeling, the stationarity of the data was examined using the Augmented Dickey-Fuller (ADF) test. Since the series were found to be non-stationary, differencing techniques were applied to achieve stationarity.

Forecasting was conducted using the AutoRegressive Integrated Moving Average (ARIMA) model. The appropriate model parameters (p, d, q) were selected using automated selection procedures. Two forecasting approaches were considered:

- A full-period model using data from 1963–2023
- A restricted model using data from 1990–2023

The restricted model was emphasized, as it better captures recent demographic trends by excluding early structural shifts. Forecasts were generated for the period 2024–2033, along with 80% and 95% confidence intervals.

Limitations of the Study

Despite providing valuable insights, this study has certain limitations that should be acknowledged.

Firstly, the analysis is based entirely on secondary data, which may be subject to reporting errors, revisions, or inconsistencies over time. The accuracy of the results depends on the reliability of the original data sources.

Secondly, the study considers only live births and crude birth rate as indicators of fertility. Other important demographic factors such as age-specific fertility rates, migration, mortality, and socio-economic variables are not included, which may limit the depth of analysis.

Thirdly, the ARIMA model is based on the assumption that past patterns will continue into the future. This assumption may not hold in the presence of sudden structural changes such as policy interventions, economic shifts, or unexpected events (e.g., pandemics).

Additionally, the presence of structural breaks in earlier decades (especially before 1990) may affect model accuracy. Although a restricted dataset was used to improve forecasting reliability, some level of distortion may still remain.

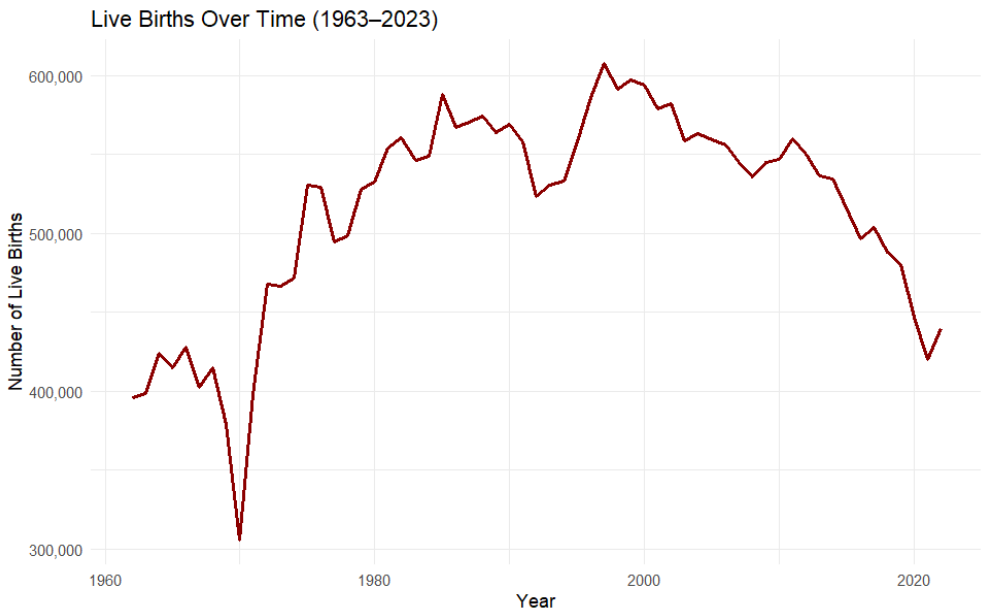
Finally, the study primarily focuses on univariate time series analysis, and does not incorporate multivariate models that could capture relationships with external explanatory variables.

DATA ANALYSIS AND RESULT

Visual Analysis

1.Live birth, crude birth and sex rate over year

Figure 1.1 illustrates the trend in the number of live births in Kerala/India between 1963 and 2023. This visualization highlights long-term demographic changes, including periods of growth, stability, and decline.

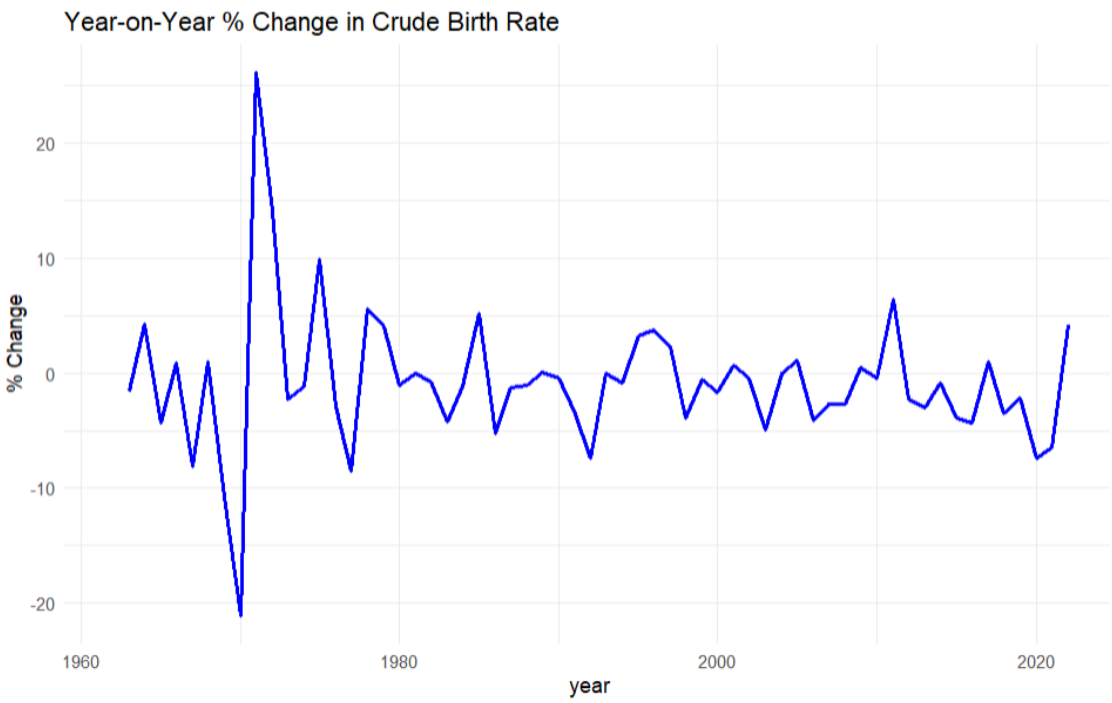


The graph shows a sharp decline in live births around 1970, followed by recovery and peaks in the late 1980s and early 1990s. Since 2000, live births have steadily declined, with a notable drop around 2020, reflecting the broader fertility transition. These fluctuations align with demographic shifts and policy interventions over the decades.

Year on Year percentage change

Year-on-year percentage change measures the annual rate of increase or decrease in live births and crude birth rate, helping to understand short-term fluctuations in demographic trends.

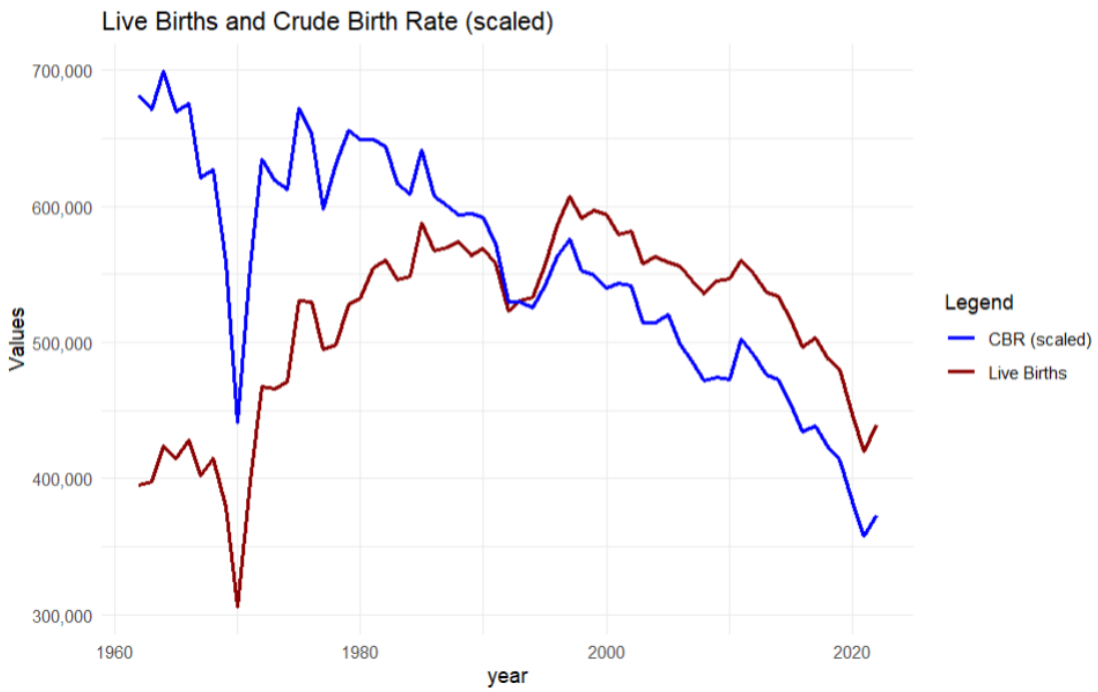
Figure 1.2 presents the year-on-year percentage change in the crude birth rate from 1963 to 2023. This visualization highlights the speed and direction of fertility changes, complementing the long-term trend analysis.



The graph shows sharp fluctuations in the early 1970s, with a steep decline followed by a sudden rebound, reflecting the impact of family planning initiatives and demographic shocks during that period. After the 1980s, the year-on-year changes became less extreme, indicating a more gradual and stable decline in fertility. This pattern underscores the transition from volatile demographic shifts to a sustained, long-term reduction in birth rates.

Live birth and Crude birth over the year from 1963 to 2023

Figure 1.3 compares the trends in live births and crude birth rate between 1963 and 2023. Live births represent the absolute number of births per year, while crude birth rate standardizes fertility relative to population size. The dual-axis plot allows simultaneous visualization of both indicators.

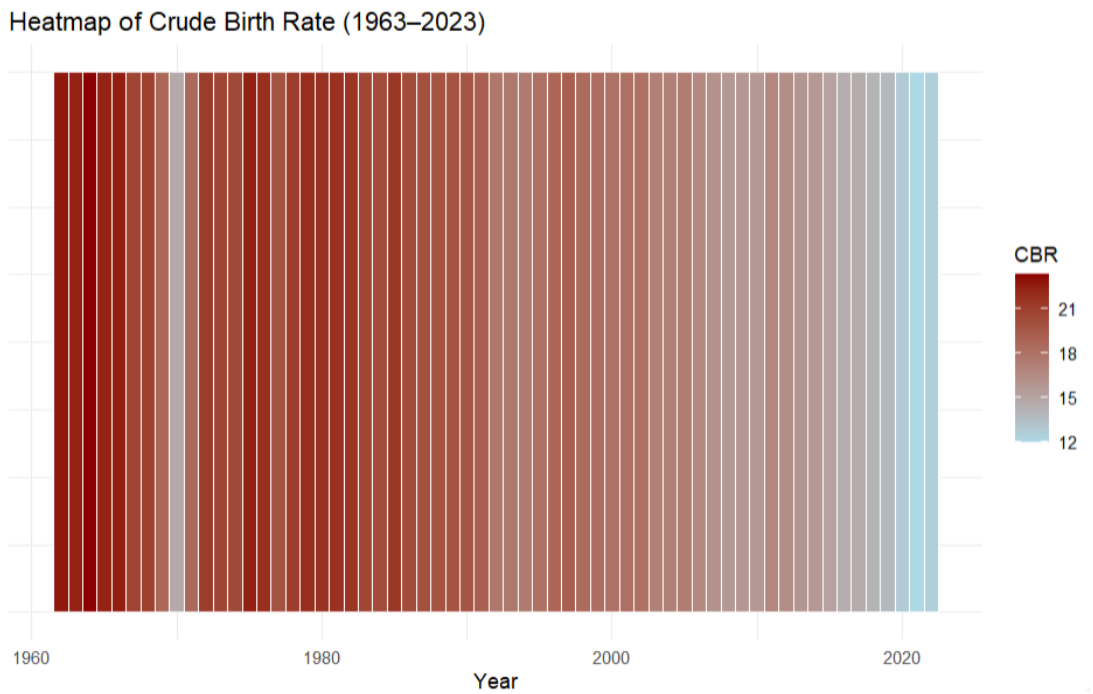


The graph shows that both live births and crude birth rate follow similar trajectories, with a sharp decline in the early 1970s, recovery in the late 1980s and early 1990s, and a steady decline after 2000. The alignment of the two measures confirms that fertility decline is not only evident in absolute numbers but also relative to population size. This reinforces the demographic transition pattern, where declining fertility accompanies broader social and economic changes.

Heatmap

A heat map converts numbers into colors so you can quickly see patterns, trends, and extremes.

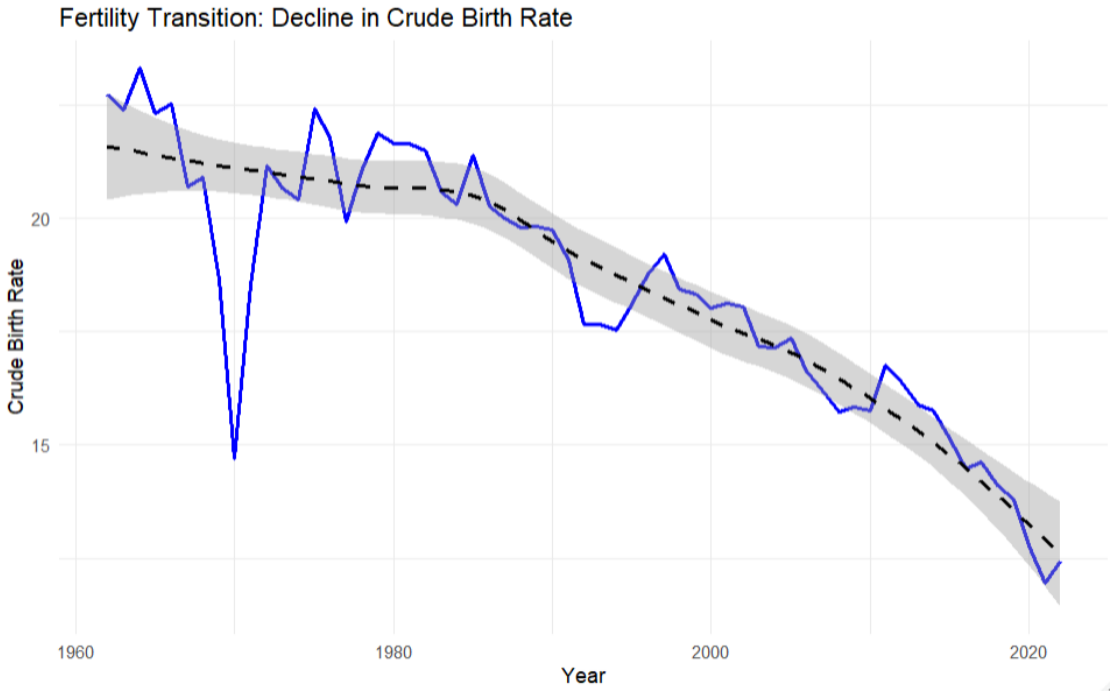
Figure 1.4 shows a heatmap of the crude birth rate from 1963 to 2023. The color gradient highlights periods of high and low fertility, with darker shades representing higher birth rates and lighter shades representing lower birth rates.



The heatmap reveals a clear fertility transition: darker shades dominate the 1960s and 1970s when crude birth rates were above 20, while lighter shades appear in the 2000s and 2010s as rates declined to around 11 by 2023. This visualization reinforces the long-term downward trend in fertility and provides an immediate visual impression of demographic change across six decades.

Fertility Transition: Decline in Crude Birth Rate

Figure 1.5 illustrates the fertility transition in Kerala/India from 1963 to 2023, showing the decline in crude birth rate over time. The smoothed trend line highlights the gradual reduction in fertility levels.



The graph shows that crude birth rate declined from around 22 in the 1960s to approximately 11 by 2023. This sustained downward trend reflects the demographic transition, where fertility reduction accompanies socioeconomic development, improved education, and family planning initiatives. The smoothed line emphasizes the long-term nature of this decline, despite short-term fluctuations.

Stationarity test result

To verify the suitability of ARIMA models, the Augmented Dickey-Fuller (ADF) test was applied to assess stationarity in the time series of live births and crude birth rate.

This table summarizes the results of the Augmented Dickey-Fuller (ADF) test applied to the time series of live births and crude birth rate.

Series	Dickey-Fuller Statistic	Lag Order	p-value	Stationarity Conclusion
Live Births	−0.88	3	0.95	Non-stationary
Crude Birth Rate	−2.62	3	0.32	Non-stationary

The results show that both series are non-stationary, with p-values greater than 0.05. This confirms the presence of long-term trends in fertility and births. Consequently, ARIMA models, which handle non-stationarity through differencing, were selected for forecasting.

Plot

Figure 1.6 presents the forecast of crude birth rate generated using the full dataset from 1963 to 2023. The forecast extends to 2033, with confidence intervals shown around the central prediction.

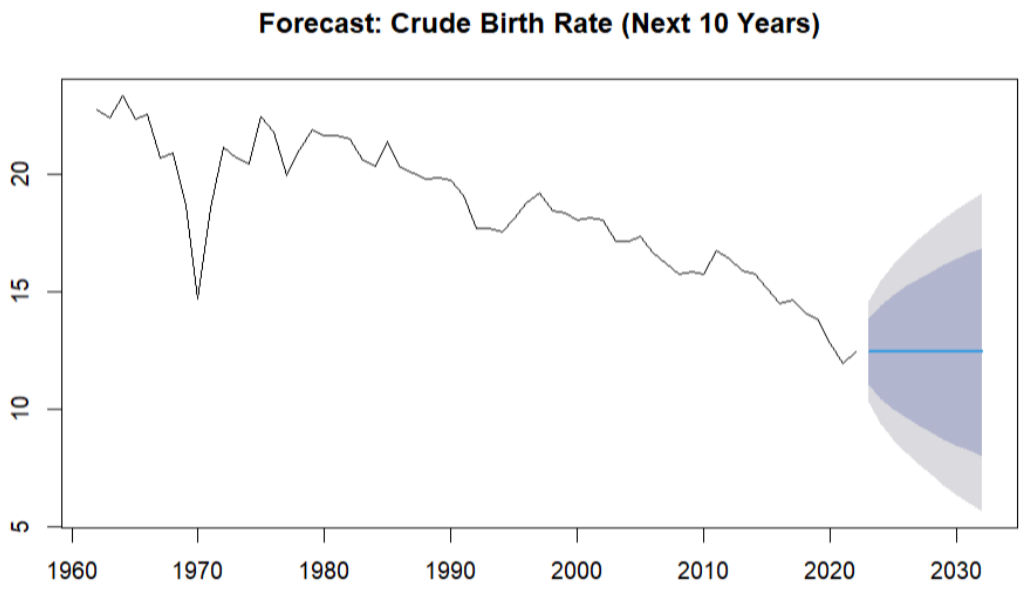


Table 1.6

Forecast of Birth Rate with 80% and 95% Pred

Year	Forecast	Lower (80%)	Upper (80%)	Lower (95%)	Upper (95%)
2023	12.43	11.03	13.83	10.29	14.57
2024	12.43	10.45	14.41	9.40	15.46
2025	12.43	10.01	14.85	8.72	16.14
2026	12.43	9.63	15.23	8.15	16.71
2027	12.43	9.30	15.56	7.64	17.22
2028	12.43	9.00	15.86	7.19	17.67
2029	12.43	8.73	16.13	6.77	18.09
2030	12.43	8.47	16.39	6.38	18.48
2031	12.43	8.23	16.63	6.01	18.85
2032	12.43	8.00	16.86	5.66	19.20

prediction Intervals (2023–2032)

While this forecast illustrates the long-term decline in fertility, it is less reliable because the inclusion of early-period data (1960s–1980s) introduces structural shifts that distort recent trends. To improve accuracy, a second forecast was generated using only the period 1990–2023, which better reflects contemporary demographic patterns. The results of that forecast are presented in Table 1.6 and Figure 1.6, and provide a more realistic projection of crude birth rate for the next decade.

*Reason for choosing crude birth rate.

n this study, crude birth rate was chosen as the primary indicator for forecasting because it provides a standardized measure of fertility relative to population size. Unlike live births, which are influenced by changes in population structure, crude birth rate reflects the underlying demographic transition more accurately. This makes CBR a more suitable measure for analyzing long-term fertility trends and for comparing across time periods and populations.

Forecast of crude birth rate by using the data from 1930-2023

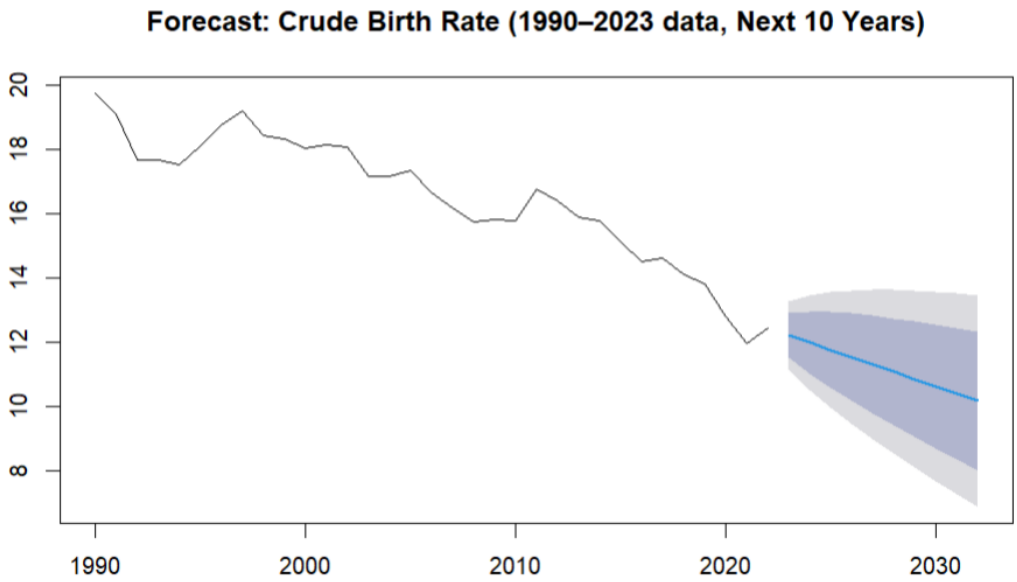
Stationarity test result

The Augmented Dickey-Fuller test confirmed that the crude birth rate series (1930–2023) is non-stationary (p-value = 0.3231). As a result, ARIMA models were applied, which handle non-stationarity through differencing. However, forecasts based on the full dataset are distorted by early structural shifts in fertility. Therefore, a restricted forecast using data from 1990–2023 was emphasized, as it better reflects contemporary demographic patterns.

Series	Dickey-Fuller Statistic	Lag Order	p-value	Conclusion
Live Births	−0.88	3	0.95	Non-stationary
Crude Birth Rate	−2.62	3	0.32	Non-stationary

Plot

Figure 1.7 presents the forecast of crude birth rate generated using ARIMA, based on data from 1990 to 2023. The forecast extends to 2033, with confidence intervals shown around the central prediction.



This forecast provides a more reliable projection compared to the full dataset forecast, as it focuses on the recent demographic transition period. By excluding early structural shifts, the model captures contemporary fertility patterns more accurately. The results indicate that crude birth rate is expected to continue its gradual decline over the next decade, consistent with ongoing demographic trends.

Table

Table summarizes the forecasted crude birth rate for the next decade (2024–2033), based on ARIMA modeling using data from 1990 to 2023. The table presents point forecasts along with 80% and 95% confidence intervals.

Year	Forecast	Lower (80%)	Upper (80%)	Lower (95%)	Upper (95%)
2024	12.20	11.52	12.88	11.16	13.24

Year	Forecast	Lower (80%)	Upper (80%)	Lower (95%)	Upper (95%)
2025	11.97	11.01	12.94	10.50	13.45
2026	11.75	10.57	12.92	9.94	13.55
2027	11.52	10.16	12.88	9.44	13.60
2028	11.29	9.77	12.81	8.96	13.62
2029	11.06	9.39	12.73	8.51	13.61
2030	10.83	9.03	12.63	8.08	13.59
2031	10.61	8.68	12.53	7.66	13.55
2032	10.38	8.34	12.42	7.26	13.50
2033	10.15	8.00	12.30	6.86	13.44

The results indicate a continued gradual decline in crude birth rate, consistent with the demographic transition observed in recent decades. By restricting the dataset to 1990–2023, the forecast avoids distortions caused by early structural shifts and provides a more accurate projection of contemporary fertility patterns.

The ARIMA model selected by auto.arima was ARIMA(0,1,0), indicating that the crude birth rate series required first differencing to achieve stationarity, but did not exhibit significant autoregressive or moving average components. This model captures the long-term declining trend in fertility without additional complexity.

In summary, the Augmented Dickey-Fuller test confirmed that both live births and crude birth rate series are non-stationary, necessitating the use of ARIMA models for forecasting. While forecasts based on the full dataset (1930–2023) illustrate the long-term decline in fertility, they are distorted by early structural shifts. By restricting the analysis to the period 1990–2023, the forecast provides a more accurate reflection of contemporary demographic trends. The results indicate that crude birth rate will continue its gradual decline over the next decade, consistent with the ongoing demographic transition in Kerala and India.

Overall, the forecast based on 1990–2023 data confirms that crude birth rate will continue its gradual decline over the next decade, reflecting the ongoing demographic transition.

Conclusion

This study examined the trends in live births and crude birth rate in Kerala over the period 1963 to 2023, highlighting the long-term demographic transition experienced by the state. The analysis revealed a clear declining trend in both live births and crude birth rate, particularly after the 1990s, reflecting improvements in education, healthcare, and family planning practices.

The year-on-year percentage change analysis showed that early decades were characterized by significant fluctuations, while recent years exhibited more stable and gradual changes. Visual tools such as heatmaps and trend plots further confirmed the steady decline in fertility levels over time.

The Augmented Dickey-Fuller test indicated that the time series data were non-stationary, necessitating the use of ARIMA models for forecasting. While forecasts based on the full dataset captured long-term trends, they were influenced by structural shifts in earlier decades. A more reliable forecast was obtained by restricting the data to the period 1990–2023, which better represents current demographic patterns.

The ARIMA model results suggest that the crude birth rate is expected to continue its gradual decline over the next decade. This finding is consistent with the ongoing demographic transition and indicates a continued movement towards lower fertility levels.

Overall, the study demonstrates that Kerala has undergone a sustained fertility decline over the past six decades, and this trend is likely to persist in the near future. These findings have important implications for population planning, resource allocation, and policy formulation.